



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Data Mining and Its Importance!



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can give information about data mining.
- I can talk about the importance of data mining in software programming.
- I can explain the concept of data mining.

Ask your partner:

- Why is it important to classify data?
- How companies keep track of their customers' account records?





Let's Explore!

Let's discuss the data mining reading!



Instructions: Read the text below.

- Mr. Adams:** Good morning, class! So, what are we reading for today's lesson?
- George:** Good morning, Mr. Adams! We're glad to see you after your surgery. Are you OK?
- Mr. Adams:** Yes, I am! Thanks for asking, I'm very happy to see you too. Also, I'm really anxious to listen to your opinions and conclusions about the reading.
- George:** Oh yes! It is about data mining. I read that it is a process that consists in examining existing databases when you need to find useful information.
- Mr. Adams:** Very good! Experts design databases for a data mining particular purpose, for example...
- George:** When you need to register scientific research, or even when you have to keep track of customers' account records. But Mr. Adams, I'm curious to know how experts perform the process of refining data in order to obtain meaningful patterns.
- Mr. Adams:** Well, first, the expert selects a set or a subset of data. Then, he classifies particular fields of interest, or what we know as variables. After that, he looks for specific patterns. One of the most common techniques to achieve this is the regression analysis technique.
- George:** Hmm, let me guess! It looks for variables that correlate with other variables, or that cluster similar data records, right?
- Mr. Adams:** Yes! Finally, the expert selects and extracts new data in a report or graph. Then, you can use it to show new information.
- George:** Excuse me, Mr. Adams! What are we studying next week?
- Mr. Adams:** Well, our next reading is about the importance of data mining.
- George:** Really? I already did it! I love this topic, I'm so happy to read about it. It explains that data mining plays an important role in businesses since it is helping companies to target different markets more every day, and online customers to find products in a faster way.
- Mr. Adams:** Exactly! Data mining of consumer information is an important application because the volume of commerce grows, and the amount of data that large systems generate, such as online bookstores and browsers increases.
- George:** I see..! When are we taking the exam about data mining?
- Mr. Adams:** Great question! You're taking it in two weeks.



Let's Practice!

George identifies what data mining is.

Instructions: Read the question. Select all the correct answers.

1. Which of the following sentences are true about data mining?
 - a. A process so important to examine existing databases to find useful information.
 - b. A study to record scientific observations and research about technology.
 - c. A practice so convenient to refine data in order to obtain meaningful patterns.
 - d. A research to find out correlated variables in the marketing field.



Let's Practice!

Data mining is evolving and providing better tools.

Instructions: Select the correct answers to complete the sentences.

1. Are you _____ a subset of data for the business next week?
 - a. identifying
 - b. identify
 - c. identified
2. Is this program finding patterns _____ to present the variables chart tomorrow morning?
 - a. yesterday night
 - b. tomorrow
 - c. tonight
3. _____ this data mining process _____ us a report this coming week?
 - a. Are / providing
 - b. Is / giving
 - c. Is / given
4. Is data mining process _____ variables tonight?
 - a. correlating
 - b. correcting
 - c. corroborating
5. _____ new and organized data for businesses in the next meeting?
 - a. You are presenting
 - b. We are presenting
 - c. Are we presenting
6. _____ we _____ particular fields of interest to structure the final report tomorrow?
 - a. Are / identifying
 - b. When / identify
 - c. When are/ identifying



Extension Activity

How much do you know about the data mining process?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about the data mining process.
- Make notes about the information you found.
- Write a short text explaining the data mining process.
- Use the grammar and vocabulary you studied in this unit.
- Read your text in front of your class.

Self-Assessment

- I identify important information about the data mining process.
- I summarize the information I found.
- I write a short text about the data mining process.
- I use the grammar and vocabulary I learned.
- I share my text with my classmates and teacher.

Yes

No

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Target Words

- cluster** (verb): to organize information according to its correlation degree
e.g. It looks for variables that correlate with other variables, or that cluster similar data records, right?
- regression analysis** (noun): the process of going back to examine something in detail
e.g. After that, he looks for specific patterns. One of the most common techniques to achieve this is the regression analysis technique.
- target** (verb): to focus on something
e.g. It is helping companies to target different markets more every day, and online customers to find products in a faster way.
- data mining** (noun): extraction of specific information from different purposes
e.g. It explains that data mining plays an important role in businesses since it is helping companies to target different markets more every day.
- correlate** (verb): to connect or to associate something
e.g. The regression analysis technique looks for variables that correlate with other variables, or that cluster similar data records.
- subset** (noun): a part of a bigger set
e.g. Well, first, the expert selects a set or a subset of data.



Learning Points

Present Continuous for Future Arrangements in Questions

You can use **the present continuous tense in interrogative form** to ask about future arrangements.

A future arrangement is something that you plan in advance and for this reason, there is a high probability to happen.

To make a statement in the present continuous tense with a future meaning, you use the same structure; only if you want to specify the future sense you can add a time complement.

Examples:

*The expert **is analyzing** data records tomorrow.*

*We **are discussing** about data mining the next Thursday.*

In this opportunity, we are going to focus on the interrogative form.

Interrogative Form Structure:

**Auxiliary verb to be in present + subject + main verb (-ing form)
+ time complement + ?**

Examples:

*Is the expert **analyzing** data records **tomorrow**?*

*Yes, he **is**.*

*No, he **is not/isn't**.*

***Are** we **discussing** about data mining **the next Thursday**?*

*Yes, we **are**.*

*No, we **are not/aren't**.*

If you want to make an information question, you need to follow this structure:

**Wh question + auxiliary verb to be in present + subject + main verb (-ing form)
+ time complement (*exception "when") + ?**

Examples:

*Why **is** the expert **analyzing** data records **tomorrow**?*

*When **are** we **discussing** about data mining?*





Learning Points

Notice that in all these questions we are asking for future plans, not about what is happening right now.

Examples

- *What are we reading for today's lesson?*
- *What are we studying next week?*
- *When are we taking the exam about data mining?*

Adjectives + Infinitives

We can use **adjectives + infinitives** in some cases to express someone's feeling about something.

Let's check first these definitions:

Adjectives:

A word that describes or modifies other word. It usually refers to physical characteristics or feelings.

For example: amazed, sad, glad, big and many others.

Infinitives:

A verb without any conjugation. You can identify it because it is always preceded by the preposition "to".

For example: to eat, to sleep, to learn, and many others.

You can make statements using the structure **adjective + infinitive** to express your opinion about something. For example:

	Adjective	Infinitive	
It is	hard	to cluster	all the data.
It seems	easy	to classify	these variables.





Learning Points

Also, to express your feelings about something. For example:

	Adjective	Infinitive	
We are	glad	to help	you with data mining.
I was	happy	to find	information really fast.

Here you find a list of frequently used adjectives which are often followed by an infinitive:

Adjectives followed by an infinitive		
Difficult	Easy	Impossible
Possible	Hard	Interesting
Boring	Safe	Unsafe
Happy	Sad	Glad
Pleased	Disappointed	Amazed

Examples

- *We are glad to see you after your surgery.*
- *I'm very happy to see you too.*
- *I'm really anxious to listen to your opinions and conclusions about the reading.*
- *I'm curious to know how experts perform the process of refining data in order to obtain meaningful patterns.*

